Peng Liu, Ph.D.

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Research Interests

Segmentation error correction for 3D volume electron microscopy (EM) organelle analysis
Label-efficient biomedical image segmentation (semi-supervised and weakly-supervised learning)
Foundation models and benchmarks for scalable multi-organelle EM segmentation

Academic Appointments and Professional Experience

Academic Appointments

Aug 2024 – Present **Boston College, Department of Computer Science** **Boston, MA, USA**
Postdoctoral Researcher (Advisor: Prof. Donglai Wei)

- Research focus: 3D volume EM segmentation for mitochondria, nuclei, and pan-organelle analysis.
- Led benchmark-centered research including MitoEM 2.0 (*Scientific Data*, under review).
- **1st Place**, CellMap Segmentation Challenge (2026; eFIB-SEM, 40+ classes).
- Ongoing work on skeleton-guided learning and multi-resolution mixture-of-experts models for instance segmentation.

Industry Experience

Jul 2017 – Jul 2019 **Meituan** **Beijing, China**
Search and Recommendation Algorithm Engineer

- Built ranking and recommendation models for travel products to improve CTR and CVR.
- Core methods: Graph Embedding, DNN, and Wide&Deep.

Education

2019 – 2024 **Shanghai Jiao Tong University** **Shanghai, China**
Ph.D. in Biomedical Engineering

- **Advisor**: Guoyan Zheng
- **Dissertation**: Semi-Supervised Medical Image Analysis Based on Consistency Regularization

2015 – 2018 **University of Chinese Academy of Sciences (Aerospace Information Research Institute)** **Beijing, China**
M.Sc. in Signal and Information Processing

- **Thesis**: Object Detection in High-Resolution Remote Sensing Images Based on Deep Learning

2011 – 2015 **China University of Geosciences** **Beijing, China**
B.Sc. in Geographic Information System

Publications ($*$ indicates equal contribution)

Journal Articles

1. **P. Liu** and G. Zheng, “Handling imbalanced data: Uncertainty-guided virtual adversarial training with batch nuclear-norm optimization for semi-supervised medical image classification,” *IEEE Journal of Biomedical and Health Informatics*, vol. 26, no. 7, pp. 2983–2994, 2022, (JCR Q1, IF=6.8).
2. **P. Liu** and G. Zheng, “Cvcl: Context-aware voxel-wise contrastive learning for label-efficient multi-organ segmentation,” *Computers in Biology and Medicine*, vol. 160, p. 106 995, 2023, (JCR Q1, IF=6.3).
3. C. Chen*, **P. Liu***, and Y. Song, “Diagnostic performance for severity grading of hip osteoarthritis and osteonecrosis of femoral head on radiographs: Deep learning model vs. board-certified orthopaedic surgeons,” *Osteoarthritis Imaging*, vol. 3, no. 2, p. 100 092, 2023.
4. **P. Liu**, M. Zhang, X. Gao, B. Li, and G. Zheng, “Joint lymphoma lesion segmentation and prognosis prediction from baseline fdg-pet images via multitask convolutional neural networks,” *IEEE Access*, vol. 10, pp. 81 612–81 623, 2022, (JCR Q2, IF=3.6).
5. Z. Li, K. Chen, **P. Liu**, X. Chen, and G. Zheng, “Entropy and distance maps-guided segmentation of articular cartilage: Data from the osteoarthritis initiative,” *International Journal of Computer Assisted Radiology and Surgery*, pp. 1–8, 2022, (JCR Q2).
6. **P. Liu** and G. Zheng, “Cr-gloco: Cross-resolution learning via global-local context consistency for semi-supervised 3d medical segmentation,” *Computerized Medical Imaging and Graphics*, 2026, Accepted and available online (in press), (JCR Q1, IF=4.9).

Preprints

1. A. Bhardwaj, C. W. Dell, M. R. Mikolaj, H. Spiers, A. Harned, B. Kuppusamy, **P. Liu**, D. Wei, E. Sterneck, and K. Narayan, “Nucleonet and dropnet: Generalist deep learning models for instance segmentation of nuclei and lipid droplets from electron microscopy images,” *bioRxiv*, p. 2026.04.02.713930, 2026. DOI: `10.64898/2026.04.02.713930`.
2. M. Wang, **P. Liu**, Y. Zhao, B. Wang, J. Wan, L. Nie, and D. Wei, “Nugraph: Graph-based reasoning over 3d primitives for nucleus segmentation correction,” *bioRxiv*, p. 2026.05.16.725603, 2026. DOI: `10.64898/2026.05.16.725603`.

Conference Proceedings

1. **P. Liu** and G. Zheng, “Semi-supervised learning regularized by adversarial perturbation and diversity maximization,” in *Machine Learning in Medical Imaging: 12th International Workshop, MLMI 2021, Held in Conjunction with MICCAI 2021*, Springer, 2021, pp. 199–208, (EI).
2. **P. Liu** and G. Zheng, “Context-aware voxel-wise contrastive learning for label efficient multi-organ segmentation,” in *International Conference on Medical Image Computing and Computer-Assisted Intervention*, Springer, 2022, pp. 653–662, (EI, CCF B).
3. Q. Zhou*, **P. Liu***, and G. Zheng, “Context-aware cutmix is all you need for universal organ and cancer segmentation,” in *Fast, Low-Resource, and Accurate Organ and Pan-Cancer Segmentation in Abdomen CT*, Springer, 2023, (EI).
4. Q. Zhou, **P. Liu**, and G. Zheng, “Partially supervised multi-organ segmentation via affinity-aware consistency learning and cross site feature alignment,” in *International Conference on Medical Image Computing and Computer-Assisted Intervention*, Springer, 2023, pp. 671–680.

Under Review

1. **P. Liu**, B. Shen, L. Liu, Q. Wang, S. Zhang, A. Bhardwaj, I. Arganda-Carreras, K. Narayan, and D. Wei, "Mitoem 2.0: A benchmark for challenging 3d mitochondria instance segmentation from em images," Major Revision at Scientific Data, 2025, (JCR Q1, IF=6.9).

Manuscripts in Preparation

1. **P. Liu** and D. Wei, *Skelaffinity-seg: Skeleton-guided geometry and global affinity learning for complex 3d tubular instance segmentation*, To be submitted to IEEE Transactions on Medical Imaging, 2026.
2. **P. Liu** and D. Wei, *Toposlot: Topology-aware structured conditional segmentation for volumetric em organelle segmentation*, To be submitted to IEEE Transactions on Medical Imaging (TMI), 2026.
3. **P. Liu** and D. Wei, *Anchorslot: Hybrid target coding for volumetric em organelle segmentation*, To be submitted to IEEE Transactions on Medical Imaging, 2026.
4. **P. Liu** and D. Wei, *Affinity-guided superpoint learning for false-merge correction in 3d em mitochondria segmentation*, To be submitted to IEEE Transactions on Medical Imaging, 2026.

Honors and Awards

Research Competitions

2026	1^{st} **Place**, CellMap Segmentation Challenge (eFIB-SEM multi-organelle segmentation, 40+ classes).
2023	2^{nd} **Place**, MICCAI FLARE 2023 Challenge (abdominal organ and pan-tumor segmentation).
2018	**Ranking: 20/120**, MDD Cup 2018 of Meituan Group. In Kaggle MDD Cup 2018.
2017	**Ranking: 7/148**, MDD Cup 2017 of Meituan Group. In Kaggle MDD Cup 2017.

Scholarships

2012	**National Encouragement Scholarship**. Awarded by Ministry of Education of China.
2012, 2013, 2014	**University Scholarships**. Awarded by China University of Geosciences.

Teaching Experience

Teaching Assistant

Shanghai Jiao Tong University, CS 2901: Programming Ideas and Methods (C), Fall 2021 and Fall 2022.

Shanghai Jiao Tong University, CS 170: Programming Ideas and Methods (C), Fall 2019 and Fall 2020.

Courses Prepared to Teach

Programming Fundamentals (C/C++/Python)

Data Structures and Algorithms

Introduction to Artificial Intelligence

Machine Learning and Deep Learning

Digital Image Processing

Medical Image Processing and Analysis

Professional Service

Journals

IEEE Transactions on Medical Imaging
Medical Image Analysis
IEEE Journal of Biomedical and Health Informatics
Neural Networks
Computerized Medical Imaging and Graphics

Conferences

MICCAI 2021
MICCAI 2022

Patents and Software

Patent

Peng Liu; Guoyan Zheng. Multi-organ segmentation method, training method, medium, and electronic device. Chinese invention patent application No. 202211185528.1, filed on September 27, 2022.

Software Copyright

Jiale Wang; Peng Liu; Guoyan Zheng. Deep-learning-based Spine Localization and Segmentation System (Spine_loc_seg) V1.0. Registration No. 2023SR0776267, registered on April 28, 2023.

Technical Skills

Languages	English (professional proficiency; TOEFL, CET-6), Mandarin Chinese (native).
Coding	Linux (7 years), C++ (10 years), Python (7 years), PyTorch (6 years), Git.